

MAE 6246: Electromechanical Control Systems

Fall 2026, F 10:00am–12:30pm, SEH 4040

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Office Hour: F 2:30–4:00pm or by appointment at SEH 3610

Course Website <https://fdcl-gwu.github.io/MAE6246/>

Course Description This course develops the state-space theory of finite-dimensional linear dynamic systems and connects mathematical analysis with computational design. Topics include state-space modeling and linearization, matrix methods, continuous- and discrete-time solutions, controllability, observability, system decompositions, stability and transient behavior, state feedback, observers, linear quadratic regulation, Kalman filtering, and LQG synthesis. Python notebooks are used throughout to analyze, simulate, and design feedback systems, with mechanical and electromechanical examples providing a common thread.

Recommended prerequisites Undergraduate linear algebra, ordinary differential equations, and introductory feedback control. Prior Python experience is helpful but not required.

Course Workload MAE 6246 is a three-credit course. Students should expect an average of 2.5 hours of direct instruction or guided interaction with the instructor and a minimum of 5 hours of independent learning each week, for a combined minimum of 7.5 hours per week.

Course Learning Objectives At the end of this course, students will be able to:

- 1: Define linear space and linear operators
- 2: Find functions of a square matrix
- 3: Find the analytic solution of linear dynamic systems
- 4: Determine controllability of linear dynamic systems
- 5: Determine observability of linear dynamic systems
- 6: Determine stability of linear dynamic systems
- 7: Design linear control systems
- 8: Design linear observers and Kalman filters
- 9: Numerically simulate controlled dynamic systems

Textbook

- C. Chen, *Linear System Theory and Design*, Sanders Publishing, 1984

Additional Readings

- B. Friedland, *Control System Design*, Dover 1986
- K. Ogata, *State Space Analysis of Control Systems*, Prentice Hall, 1967
- W. Brogan, *Modern Control Theory*, Prentice Hall, 1991
- T. Kailath, *Linear Systems*, Prentice Hall, 1980

Grading Homework 37.5%, Attendance 2.5%, Midterm Exam 30% (October 16th), Final Exam 30%

Course Policy

- 1: **Check your GW email account daily.** All of the important announcements of this class will be made through your email.
- 2: **Homework:** There is a **late penalty of 1 pt per 10 minutes**. The late penalty is enforced strictly except under extraordinary circumstances such as documented illness or emergency. Student must inform the instructor in writing with supporting documents **before** the due date for waiver.
- 3: **Exam:** Make-up exams will not be given, except under extraordinary circumstances such as documented illness or emergency. Student must inform the instructor in writing with supporting documents at least **a week** ahead of time to be allowed for a make up exam. If an exam is simply missed, then a grade of zero will be recorded.
- 4: **Generative AI:** You may use generative AI tools for assignments, provided that:
 - Representative queries and responses that materially influenced your solution are exported to PDF and should be submitted as supplementary material;
 - You may be asked to explain, modify, or extend portions of the submitted work at any time to assess your understanding;
 - You are fully responsible for the correctness, originality, and interpretation of the submitted work.

This policy implements the University's August 2026 guidance:

<https://provost.gwu.edu/guidelines-using-artificial-intelligence>.

University Policies

- 1: **Academic Integrity Code:** Academic integrity is an essential part of the educational process, and all members of the GW community take these matters very seriously. As the instructor of record for this course, my role is to provide clear expectations and uphold them in all assessments. Violations of academic integrity occur when students fail to cite research sources properly, engage in unauthorized collaboration, falsify data, and otherwise violate the Code of Academic Integrity. If you have any questions about whether particular academic practices or resources are permitted, you should ask me for clarification. If you are reported for an academic integrity violation, you should contact Conflict Education and Student Accountability (CESA), formerly known as Student Rights and Responsibilities (SRR), to learn more about your rights and options in the process. Consequences can range from failure of assignment to expulsion from the University and may include a transcript notation. For more information, refer to the CESA website at <https://students.gwu.edu/code-academic-integrity> or contact CESA by email cesa@gwu.edu or phone 202-994-6757.
- 2: **University Policy on Observance of Religious Holidays:** Students must notify faculty during the first week of the semester in which they are enrolled in the course, or as early as possible, but no later than three weeks prior to the absence, of their intention to be absent from class on their day(s) of religious observance. If the holiday falls within the first three weeks of class, the student must inform faculty in the first week of the semester. For details and policy, see <https://provost.gwu.edu/policies-procedures-and-guidelines>.
- 3: **Use of Electronic Course Materials and Class Recordings:** Students are encouraged to use electronic course materials, including recorded class sessions, for private personal use in connection with their academic program of study. Electronic course materials and recorded class sessions should not be shared or used for non-course related purposes unless express permission has been granted by the instructor. Students who impermissibly share any electronic course materials are subject to discipline under the Student Code of Conduct.

Contact the instructor if you have questions regarding what constitutes permissible or impermissible use of electronic course materials and/or recorded class sessions. Contact Disability Support Services at <https://disabilitysupport.gwu.edu> if you have questions or need assistance in accessing electronic course materials.

Academic Support

- 1: **Academic Commons:** Academic Commons is the central location for academic support resources for GW students. To schedule a peer tutoring session for a variety of courses visit <https://go.gwu.edu/tutoring>. Visit <https://academiccommons.gwu.edu> for study skills tips, finding help with research, and connecting with other campus resources. For questions email academiccommons@gwu.edu.
- 2: **GW Writing Center:** GW Writing Center cultivates confident writers in the University community by facilitating collaborative, critical, and inclusive conversations at all stages of the writing process. Working alongside peer mentors, writers develop strategies to write independently in academic and public settings. Appointments can be booked online at <https://writingcenter.gwu.edu>.

Support for Students In and Outside the Classroom

- 1: **Disability Support Services (DSS) 202-994-8250:** Any student who may need an accommodation based on the potential impact of a disability should contact Disability Support Services at <https://disabilitysupport.gwu.edu> to establish eligibility and to coordinate reasonable accommodations.
- 2: **Office of Advocacy and Support (OAS):** The Office of Advocacy and Support (OAS) is the university's sole dedicated confidential support resource for student survivors of sexual assault, domestic violence, and stalking. OAS provides trauma-informed and healing-centered emotional support, safety planning, academic and logistical support, and assistance with on and off-campus resource navigation and referrals. Connecting with OAS does not result in reporting to the institution. OAS can be reached at oas@gwu.edu or (202) 994-0443.
- 3: **Student Health Center (SHC):** The SHC offers medical, counseling/psychological, and psychiatric services to GW students and can be reached at 202-994-5300. More information is available on the Student Health Center website. Students experiencing a medical or mental health emergency on campus should contact GW Emergency Services at 202-994-6111, or off campus at 911.

GW Campus Emergency Information

- 1: **GW Emergency Services: 202-994-6111:** For situation-specific instructions, refer to GW's Emergency Procedures guide.
- 2: **GW Alert:** GW Alert is an emergency notification system that sends alerts to the GW community. GW requests students, faculty, and staff maintain current contact information by logging on to <https://alert.gwu.edu>. Alerts are sent via email, text, social media, and other means, including the Guardian app. The Guardian app is a safety app that allows you to communicate quickly with GW Emergency Services, 911, and other resources. Learn more at <https://safety.gwu.edu>.
- 3: **Protective Actions:** GW prescribes four protective actions that can be issued by university officials depending on the type of emergency. All GW community members are expected to follow directions according to the specified protective action. The protective actions are Shelter, Evacuate, Secure, and Lockdown. Learn more at <https://safety.gwu.edu/gw-standard-emergency-statuses>.

Course Calendar The schedule may be adjusted to match the pace of the class. Blackboard announcements and the course website will identify any changes.

Week	Date	Topic	Representative Python work
1	Aug. 28	State-space models and linearization	Nonlinear pendulum; linearization about downward and upright equilibria
2	Sept. 4	Linear operators and computational linear algebra	Eigenstructure, SVD, conditioning, modal coordinates, and matrix exponential
3	Sept. 11	Solutions of continuous-time linear systems	Compare matrix exponential, numerical integration, and input-response simulation
4	Sept. 18	Sampling, discrete-time models, and controllability	Discretize a DC motor; controllability tests and transfer energy
5	Sept. 25	Observability and state reconstruction	Compare sensor choices and reconstruction conditioning
6	Oct. 2	Canonical decompositions and minimal realization	Decompose a model and verify input-output equivalence
7	Oct. 9	Stability and transient behavior	Lyapunov equation; stable non-normal systems with transient amplification
8	Oct. 16	Midterm examination	Coverage is limited to Weeks 1–6
9	Oct. 23	State feedback and pole placement	Inverted-pendulum stabilization, tracking, and integral action
10	Oct. 30	State observers and output feedback	Observer-based control with measurement noise
11	Nov. 6	Optimal-control foundations and finite-horizon LQR	Double-integrator regulation and the state-effort tradeoff
12	Nov. 13	Riccati equations and infinite-horizon LQR	Continuous/discrete LQR and systematic weight sweeps
13	Nov. 20	Kalman filtering	Estimate position and velocity from noisy measurements
–	Nov. 27	Thanksgiving break – no class	–
14	Dec. 4	LQG synthesis and course integration	Integrated control and estimation with model mismatch and saturation